

Thermo-Hydraulic Characterization of Non-Newtonian and Hemorheological Flows in Branched Microchannels

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1. Introduction

The study of non-Newtonian fluid flow coupled with convective heat transfer within microfluidic channels and complex conduits has emerged as a critical domain across biomedical engineering, targeted drug delivery, and microscale thermal management systems [1, 2]. Biological fluids, most notably human blood flowing through microvascular networks or localized constrictions, display pronounced non-Newtonian rheological behavior where apparent viscosity varies nonlinearly with the applied strain rate [3, 4]. Capturing these constitutive flow dynamics requires robust mathematical representations such as the Ostwald–de Waele power law model where the power-law flow behavior index (n) dictates whether the fluid exhibits shear-thinning ($n < 1$), Newtonian ($n = 1$), or shear-thickening ($n > 1$) characteristics [5, 6]. Concurrently, temperature-dependent viscosity variations and localized thermal gradients induce substantial shifts in velocity distribution, pressure drop, and wall shear stress, thereby governing the overall thermo-hydraulic performance of the system [7, 8].

Understanding the interplay between variable fluid rheology and imposed thermal boundary conditions has driven considerable computational and experimental research. Nejad et al. [9] and Gijzen et al. [10] emphasized the physiological importance of non-Newtonian modeling in confined vascular and stenosed geometries, demonstrating that shear-thinning effects alter velocity profiles and reduce resistance in high-shear regions. In microchannel and heat sink applications, non-Newtonian carriers are frequently implemented to enhance convective heat transfer rates while optimizing pumping power requirements [11, 12]. Furthermore, thermal gradients across channel walls give rise to localized thermophysical property variations, where even slight shifts in temperature differences (ΔT) modify near-wall viscosity gradients and thermal boundary layer development [13, 14].

Despite substantial advancements in microfluidic and rheological modeling, existing literature frequently examines non-Newtonian flow under isothermal conditions or assumes fixed fluid indices under standard thermal environments. Systematic investigations exploring the coupled effects of varying power-law indices across both pseudoplastic and dilatant regimes under progressively varying thermal gradients remain insufficiently documented.

To bridge this gap, the present study conducts a comprehensive finite element numerical investigation using COMSOL Multiphysics to examine the thermo-hydraulic characteristics of non-Newtonian fluid flow across a parametric sweep of the flow behavior index ($n = 0.7$ to 1.3) under systematically controlled wall temperature differentials. The imposed baseline temperature is maintained at $T_0 = 310$ K, while the heating boundary temperature is varied incrementally ($T_w = 315$ K, 313 K, and 311 K), establishing temperature differences of $\Delta T = 5$ K, 3 K, and

1 K. Parametric sweeps are executed to evaluate the velocity fields, pressure distributions, apparent viscosity maps, and local heat transfer rates across both shear-thinning ($n < 1$) and shear-thickening ($n > 1$) regimes. This investigation provides detailed physical insights into optimizing fluid transport and thermal management in biomedical and microchannel architectures.

2. Mathematical Formulation and Computational Methodology

2.1 Geometric Configuration and Domain Description

The computational domain represents a two-dimensional branched microchannel geometry designed to capture convective transport and fluid deformation in complex flow junctions. As defined in the numerical domain, the configuration comprises a primary horizontal microchannel segment of length $L_1 = 3.5W$ and height $W = 1$ m, integrated with symmetric angled branching arms characterized by a branch length $L_2 = 2W$. The overall flow domain is constructed via boolean union operations of planar rectangular subdomains and transitional connecting polygonal sections. The geometric boundaries comprise an upstream inflow boundary, an exit outflow boundary, and stationary solid bounding walls.

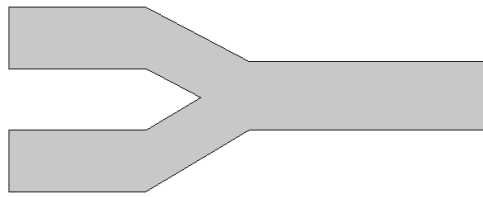


Figure 1: Geometry of the bifurcated channel

2.2 Governing Equations

The transport phenomena governing the laminar, time-dependent, incompressible non-Newtonian flow coupled with convective heat transfer are formulated through the conservation of mass, momentum, and energy:

- **Conservation of Mass (Continuity Equation):**

$$\nabla \cdot \mathbf{u} = 0$$

where $\mathbf{u} = (u, v)$ denotes the two-dimensional velocity vector field.

- **Conservation of Momentum (Navier–Stokes Equation):**

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \nabla \cdot \boldsymbol{\tau}$$

where ρ represents the fluid density (1000 kg/m³), p is the static pressure field, and $\boldsymbol{\tau}$ is the viscous extra stress tensor defined as:

$$\boldsymbol{\tau} = 2\mu_{app} \mathbf{S} = \mu_{app} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$$

where $\mathbf{S} = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$ is the rate-of-strain tensor.

- **Non-Newtonian Constitutive Relationship (Ostwald–de Waele Power-Law Model):**

The apparent dynamic viscosity μ_{app} of the fluid varies nonlinearly as a function of the local shear rate magnitude according to the power-law formulation:

$$\mu_{app} = m \cdot \left(\frac{\dot{\gamma}}{\dot{\gamma}_{ref}} \right)^{n-1}$$

where m denotes the fluid consistency index ($m = 10$ s), $\dot{\gamma}_{ref} = 1 \text{ s}^{-1}$ is the reference shear rate, and n is the dimensionless flow behavior index. The scalar shear rate $\dot{\gamma}$ is calculated from the second invariant of the rate-of-strain tensor:

$$\dot{\gamma} = \sqrt{2\mathbf{S}:\mathbf{S}} = \sqrt{2 \sum_{i,j} S_{ij} S_{ij}}$$

The power-law index is systematically varied across seven discrete values ($n = 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3$) to encompass shear-thinning (pseudoplastic, $n < 1.0$), standard Newtonian ($n = 1.0$), and shear-thickening (dilatant, $n > 1.0$) flow regimes.

- **Conservation of Energy (Thermal Convection-Diffusion Equation):**

$$\rho C_p \left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) = \nabla \cdot (k \nabla T)$$

where C_p is the specific heat capacity, k is the thermal conductivity, and T represents the local temperature field.

2.3 Boundary and Initial Conditions

The boundary and initial conditions imposed across the computational domain are defined as follows:

- Inlet Boundary Condition: A fully developed laminar parabolic velocity profile is imposed at the channel entrance ($x = 0$):

$$u(y) = 4(y - y^2), \quad v = 0$$

- Outlet Boundary Condition: A constant gauge static pressure is specified at the downstream outlet:

$$p = 0 \text{ Pa}, \quad \nabla \mathbf{u} \cdot \mathbf{n} = 0$$

with a supplementary pressure point constraint applied to fix the pressure reference.

- Solid Wall Boundary Condition: The standard no-slip and no-penetration conditions are enforced along all rigid solid channel boundaries:

$$u = 0, \quad v = 0$$

- Thermal Boundary Conditions: A baseline reference temperature is assigned at the inlet and unheated boundaries ($T_0 = 310 \text{ K}$). A constant elevated temperature is applied to the active heated wall sections, parameterized across three thermal differentials:

$$T_w = 315 \text{ K} \quad (\Delta T = 5 \text{ K})$$

$$T_w = 313 \text{ K} \quad (\Delta T = 3 \text{ K})$$

$$T_w = 311 \text{ K} \quad (\Delta T = 1 \text{ K})$$

- Initial Conditions ($t = 0 \text{ s}$): The domain starts with the parabolic velocity field $u(x, y, 0) = 4(y - y^2)$, $v(x, y, 0) = 0$, uniform initial gauge pressure $p(x, y, 0) = 0 \text{ Pa}$, and uniform temperature $T(x, y, 0) = 310 \text{ K}$.

2.4 Numerical Implementation and Solver Schemes

The coupled partial differential equations are discretized and solved numerically using the Finite Element Method (FEM) in COMSOL Multiphysics. Spatial discretization is executed over an unstructured free triangular mesh, with a maximum element size controlled by the characteristic mesh parameter ($h_{max} = 0.01 \text{ m}$).

Time integration is advanced over the interval $t = 0$ to 5 s with a discrete step size of $\Delta t = 0.1 \text{ s}$ using the Backward Differentiation Formula (BDF) solver. The resulting non-symmetric linearized system of algebraic equations is solved using the direct parallel sparse solver (PARDISO). Convergence at each time step is enforced with a relative tolerance of 0.005. Fully

coupled nonlinear Newton iterations are accelerated using Anderson acceleration to ensure numerical stability and robust convergence throughout the transient cycle.

3. Results and Discussion

The finite element numerical simulations evaluate the coupled hydrodynamic and thermal transport phenomena across a spectrum of power-law indices ($n = 0.7$ to 1.3) under discrete wall temperature differentials ($\Delta T = 5$ K, 3 K, and 1 K). The resulting flow dynamics, viscous dissipation, pressure distributions, and convective heat transfer characteristics are analyzed across transient time instances from $t = 0$ s to $t = 5$ s.

3.1 Influence of Power-Law Index (n) on Velocity Field and Flow Dynamics

The velocity magnitude distribution across the microchannel geometry illustrates a profound dependence on the non-Newtonian flow behavior index n .

- For pseudoplastic or shear-thinning fluid regimes ($n < 1.0$), the apparent viscosity diminishes near the channel boundaries where high strain rates ($\dot{\gamma}$) prevail. This localized viscosity reduction suppresses near-wall viscous resistance, yielding a flattened, plug-like velocity profile in the central core while steepening the velocity gradient adjacent to the solid walls.
- For Newtonian flow ($n = 1.0$), the velocity profile preserves its classical parabolic Poiseuille shape throughout the straight channel before dividing smoothly into the angled branches.
- For dilatant or shear-thickening fluids ($n > 1.0$), high shear rates near the walls induce elevated apparent viscosity, restricting fluid motion near the solid boundaries. Consequently, fluid momentum is channeled into an accelerated central jet toward the junction apex, resulting in elevated core velocity peaks and stronger shear strain concentrations at the branching divider.

3.2 Static Pressure Field and Hydrodynamic Resistance

The static pressure field development demonstrates the energetic penalty associated with shifting fluid rheology.

- As the power-law exponent increases from $n = 0.7$ to $n = 1.3$, the overall pressure drop ($\Delta P = P_{\text{inlet}} - P_{\text{outlet}}$) across the microchannel junction increases monotonically.

- In shear-thinning fluids ($n = 0.7$), the reduced wall shear stress results in gentle axial pressure gradients and minimal energetic loss during stream splitting.
- In contrast, shear-thickening fluids ($n = 1.3$) exhibit substantial viscous resistance, producing steep upstream pressure gradients required to sustain the prescribed inflow rate through the branching arms. Over transient time progression from $t = 1$ s to $t = 5$ s, the pressure field stabilizes into a steady convective distribution across all cases.

3.3 Thermal Convection and Temperature Field Evolution across Thermal Differentials

Thermal transport is governed by the strong coupling between non-Newtonian velocity boundary layers and imposed wall temperature differentials (ΔT). The thermal performance across the microchannel geometry is evaluated for three discrete wall temperature differentials relative to the baseline inlet temperature $T_0 = 310$ K: $\Delta T = 1$ K ($T_w = 311$ K), $\Delta T = 3$ K ($T_w = 313$ K), and $\Delta T = 5$ K ($T_w = 315$ K).

- Influence of Temperature Differential Magnitude (ΔT): Scaling the temperature differential from 1 K to 5 K increases the transverse thermal driving potential without altering the qualitative shape of the isotherms. For the minimal differential ($\Delta T = 1$ K), thermal diffusion is confined tightly to the immediate vicinity of the heated walls, and the core fluid travels through the bifurcation apex near the undisturbed baseline temperature of 310 K. Conversely, at $\Delta T = 5$ K, significant thermal boundary layer growth occurs, establishing steep transverse temperature gradients and promoting deeper thermal penetration into the bulk flow.
- Rheological Modulation ($n = 0.7$ vs. 1.0 vs. 1.3): Across all temperature levels, the power-law flow behavior index dictates thermal advection efficiency. In shear-thinning fluids ($n = 0.7$), elevated near-wall fluid velocity significantly augments convective heat advection, driving heat deeper into the core stream. In shear-thickening fluids ($n = 1.3$), localized viscosity buildup at the walls retards near-wall fluid motion, suppressing convective diffusion and trapping sharp temperature gradients in a narrow near-wall band. For the Newtonian fluid ($n = 1.0$), thermal layer growth maintains an intermediate, classical parabolic profile.

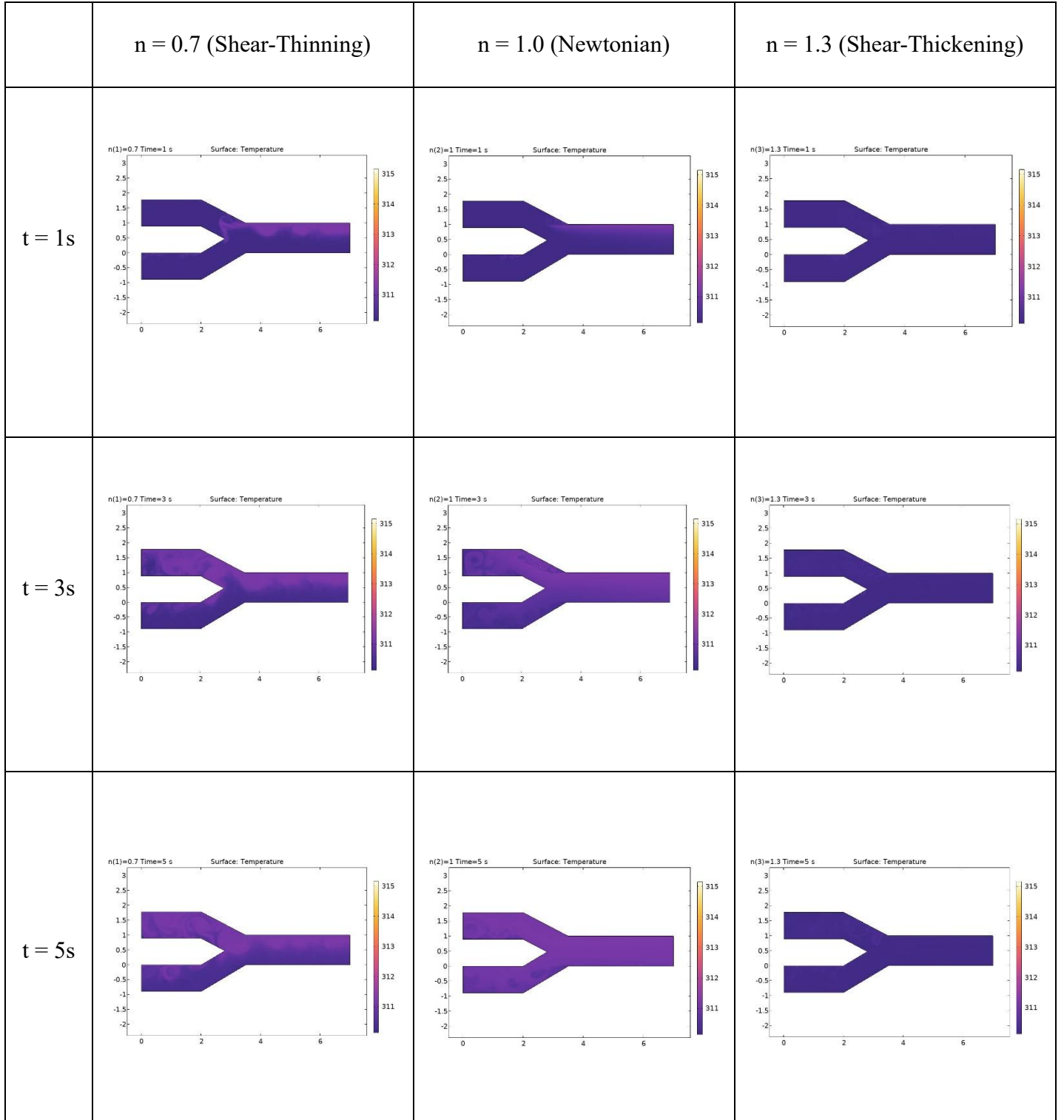


Fig. 2. Spatiotemporal temperature contours for $\Delta T = 1\text{ K}$ ($T_w = 311\text{ K}$) across $n = 0.7, 1.0, 1.3$ at $t = 1\text{ s}, 3\text{ s}, 5\text{ s}$.

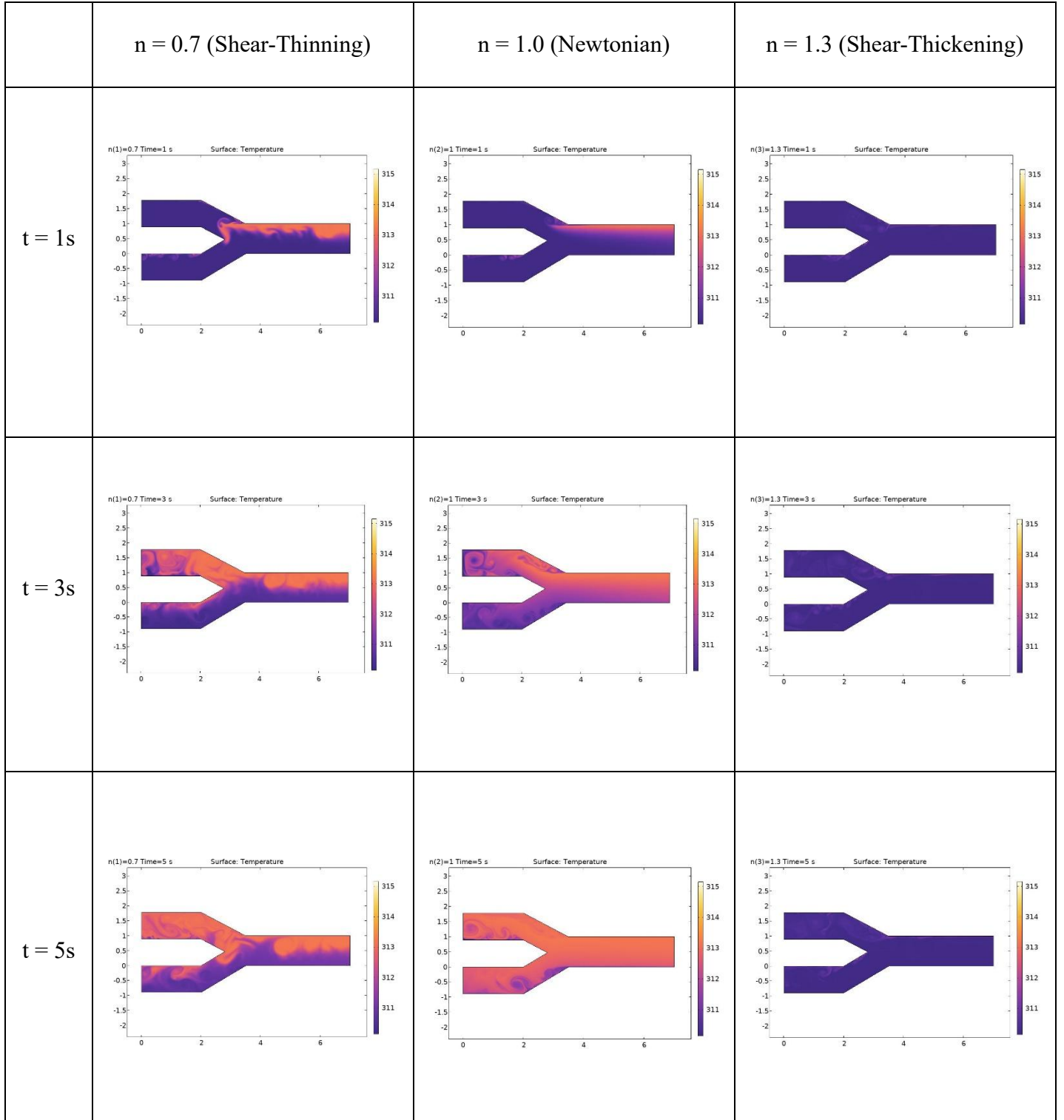


Fig. 3. Spatiotemporal temperature contours for $\Delta T = 3\text{ K}$ ($T_w = 313\text{ K}$) across $n = 0.7, 1.0, 1.3$ at $t = 1\text{ s}, 3\text{ s}, 5\text{ s}$.

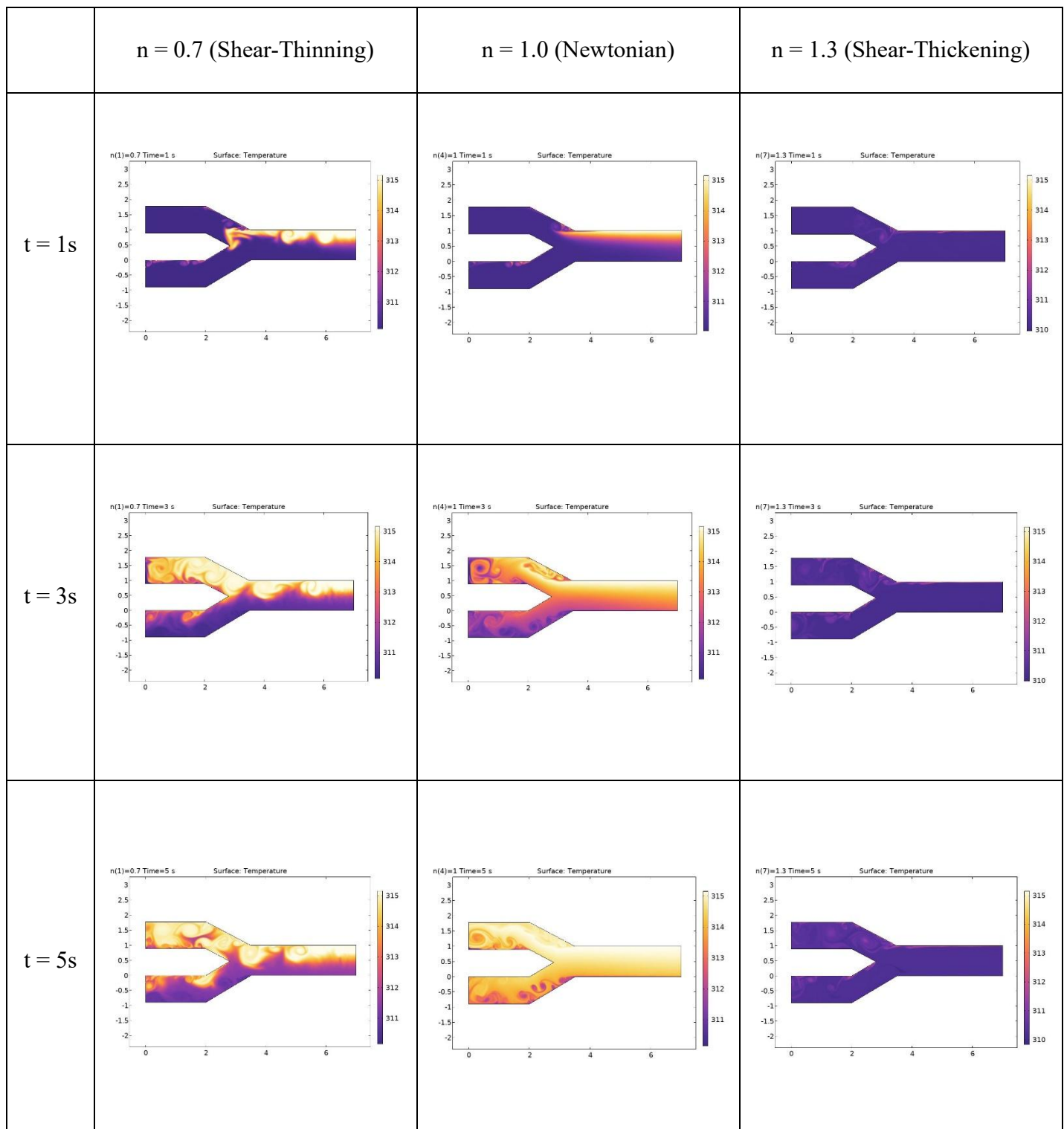


Fig. 4. Spatiotemporal temperature contours for $\Delta T = 5\text{ K}$ ($T_w = 315\text{ K}$) across $n = 0.7, 1.0, 1.3$ at $t = 1\text{ s}, 3\text{ s}, 5\text{ s}$.

3.2 Static Pressure Field and Hydrodynamic Resistance

The static pressure distribution reflects the energetic requirements of forcing non-Newtonian fluids through the bifurcating microchannel junction.

- Rheological Dependence: As the power-law exponent increases from $n = 0.7$ to $n = 1.3$, the overall pressure drop ($\Delta P = P_{\text{inlet}} - P_{\text{outlet}}$) across the channel increases monotonically.
- Energy Dissipation in Pseudoplastic vs. Dilatant Regimes: In shear-thinning fluids ($n = 0.7$), reduced apparent viscosity in regions of high strain rate minimizes wall shear resistance, producing low static pressure penalties across the branching arms. Conversely, in shear-thickening fluids ($n = 1.3$), elevated viscous resistance generates steep upstream axial pressure gradients to sustain throughput, yielding high pumping requirements. Over the transient duration ($t = 1$ s to 5 s), localized pressure contours stabilize into a steady distribution across all test cases.

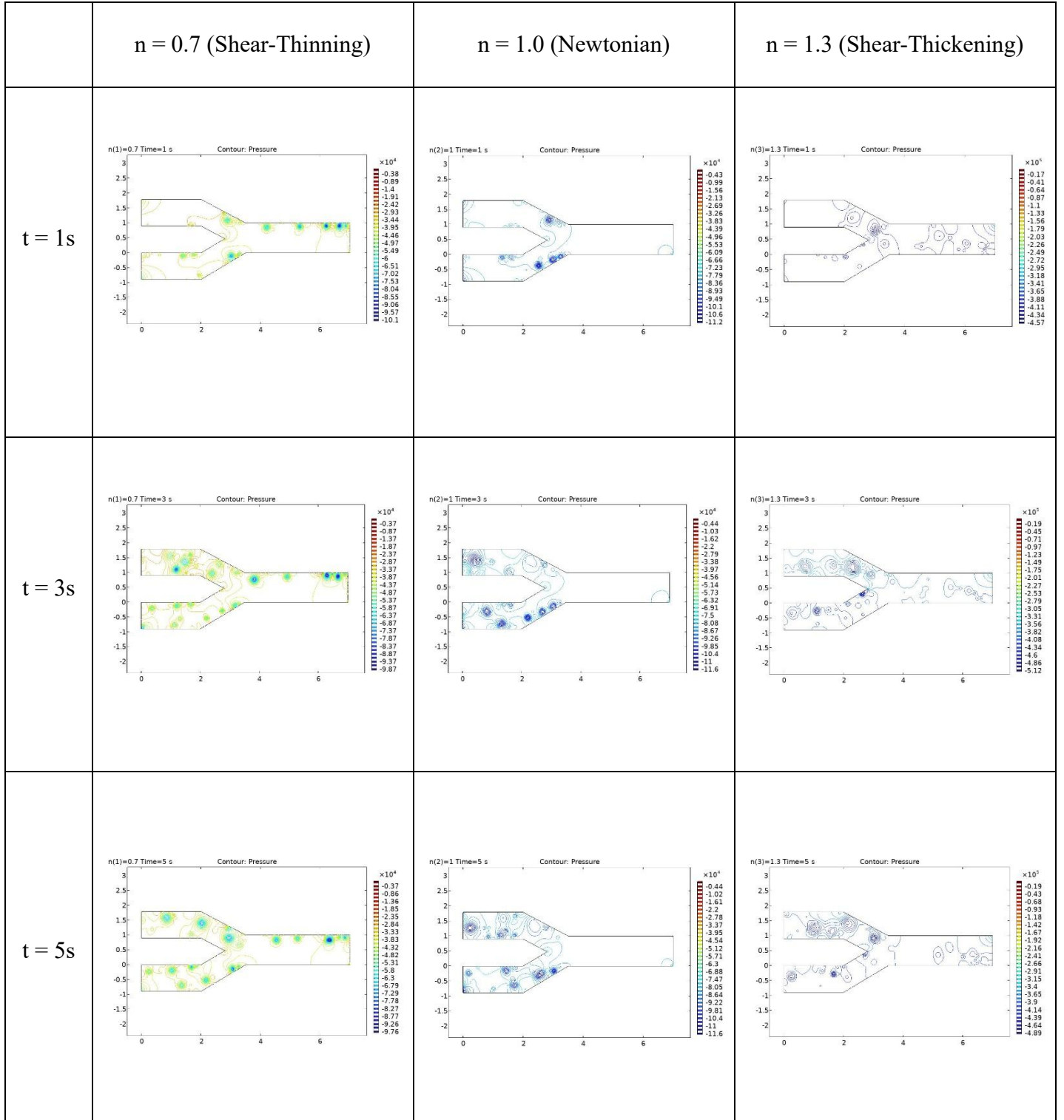


Fig. 5. Pressure distribution contour patterns across the branching junction for $\Delta T = 1$ K ($T_w = 311$ K) across $n = 0.7, 1.0, 1.3$ at $t = 1$ s, 3 s, 5 s.

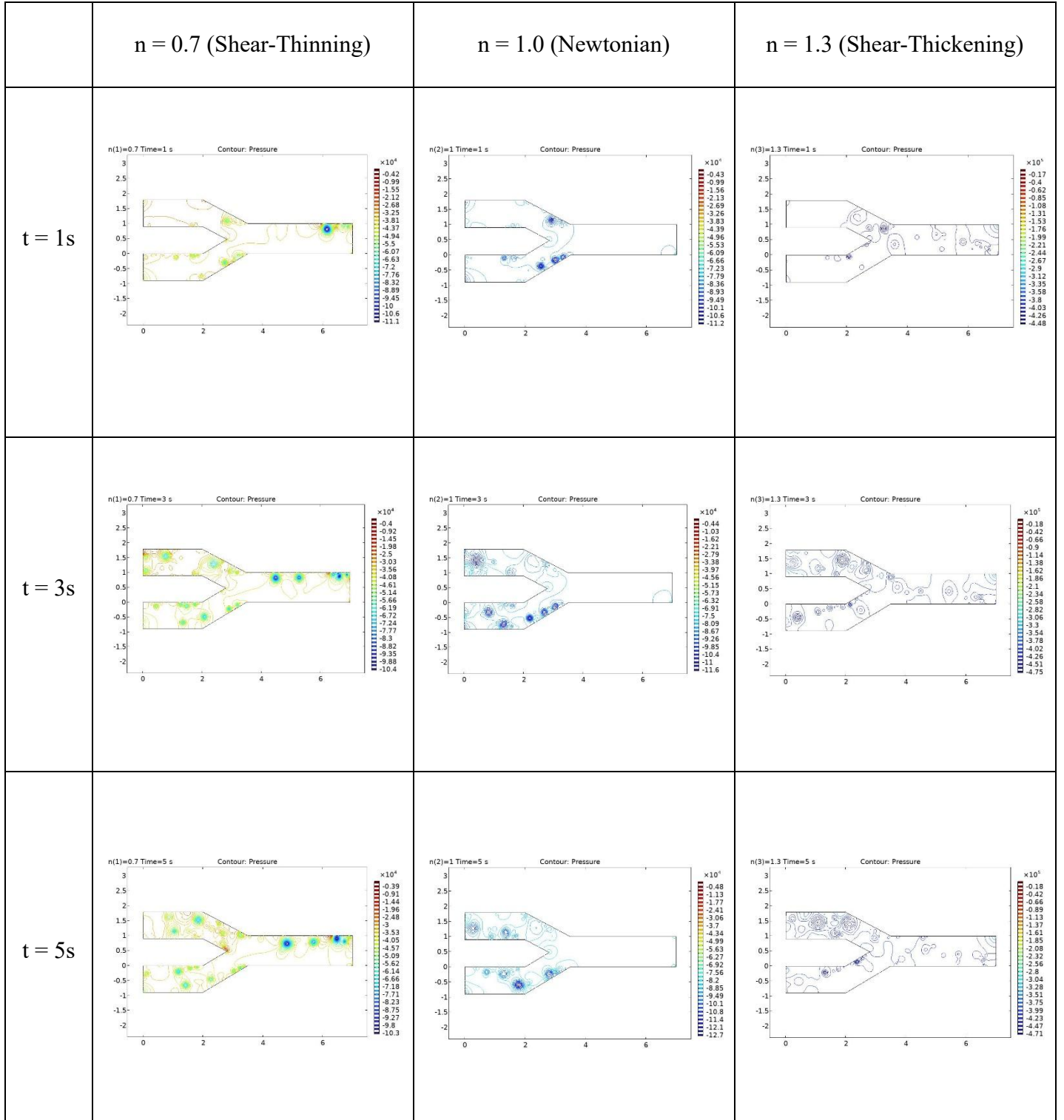


Fig. 6. Pressure distribution contour patterns across the branching junction for $\Delta T = 3$ K ($T_w = 313$ K) across $n = 0.7, 1.0, 1.3$ at $t = 1$ s, 3 s, 5 s.

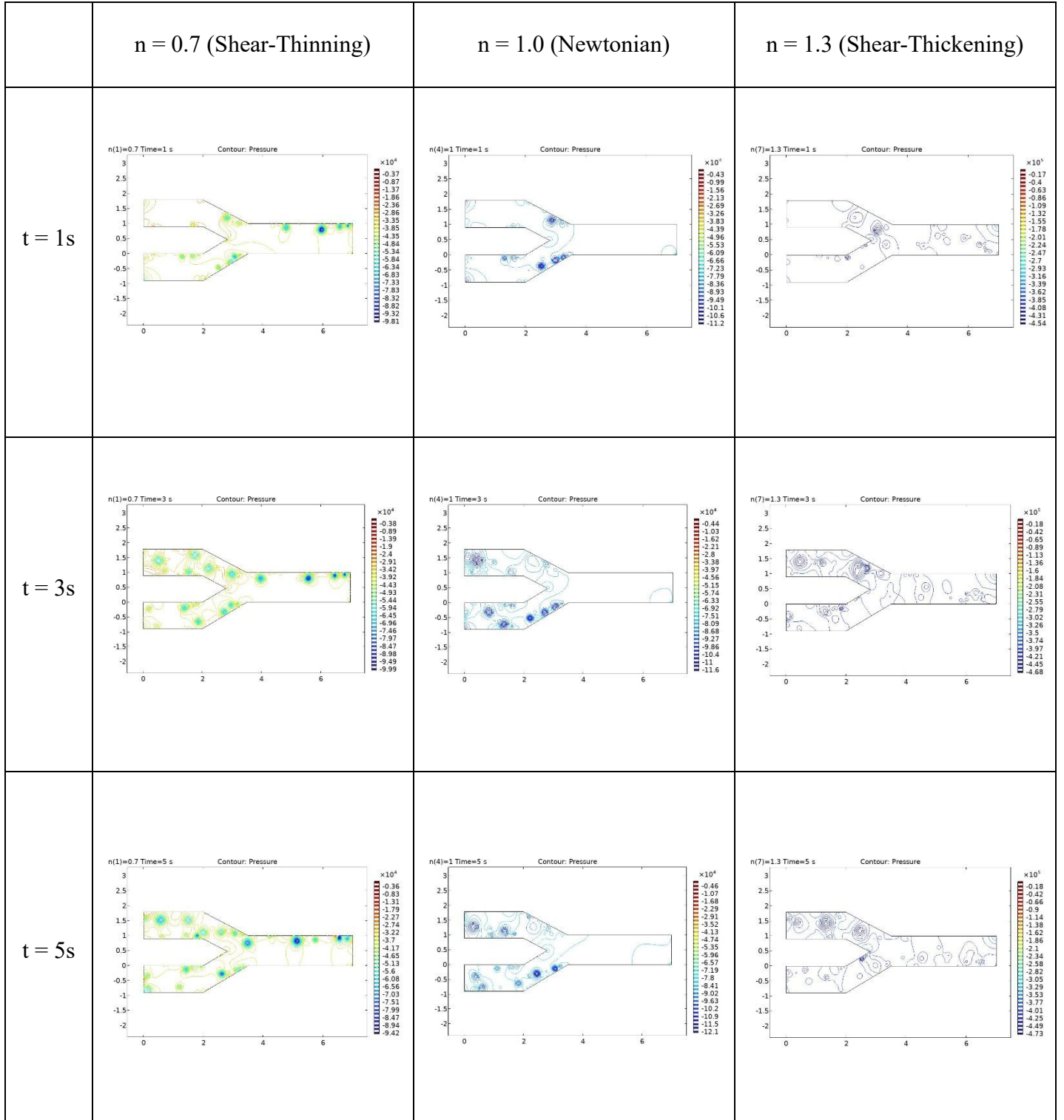


Fig. 7. Pressure distribution contour patterns across the branching junction for $\Delta T = 5$ K ($T_w = 315$ K) across $n = 0.7, 1.0, 1.3$ at $t = 1$ s, 3 s, 5 s.

3.3 Velocity Field and Flow Dynamics

The velocity magnitude distribution illustrates the kinematic response of the fluid as it negotiates the dividing apex of the branched channel.

- **Velocity Profiles:** For shear-thinning fluids ($n < 1.0$), localized viscosity reduction adjacent to the walls suppresses viscous drag, resulting in a blunt, plug-like velocity profile in the central lumen and steep velocity gradients at the boundary.
- **Newtonian and Dilatant Behavior:** For $n = 1.0$, the velocity profile preserves its parabolic Poiseuille shape. For shear-thickening fluids ($n > 1.0$), high apparent viscosity along the channel walls constricts the boundary flow, accelerating the central fluid core into an intensive jet directly toward the bifurcation divider.
- **Creeping Flow Behavior:** Consistent with low-Reynolds-number microscale flow ($Re = 10^{-2}$), fluid streams divide symmetrically into upper and lower branching arms without inducing secondary flow separation or wake vortices.

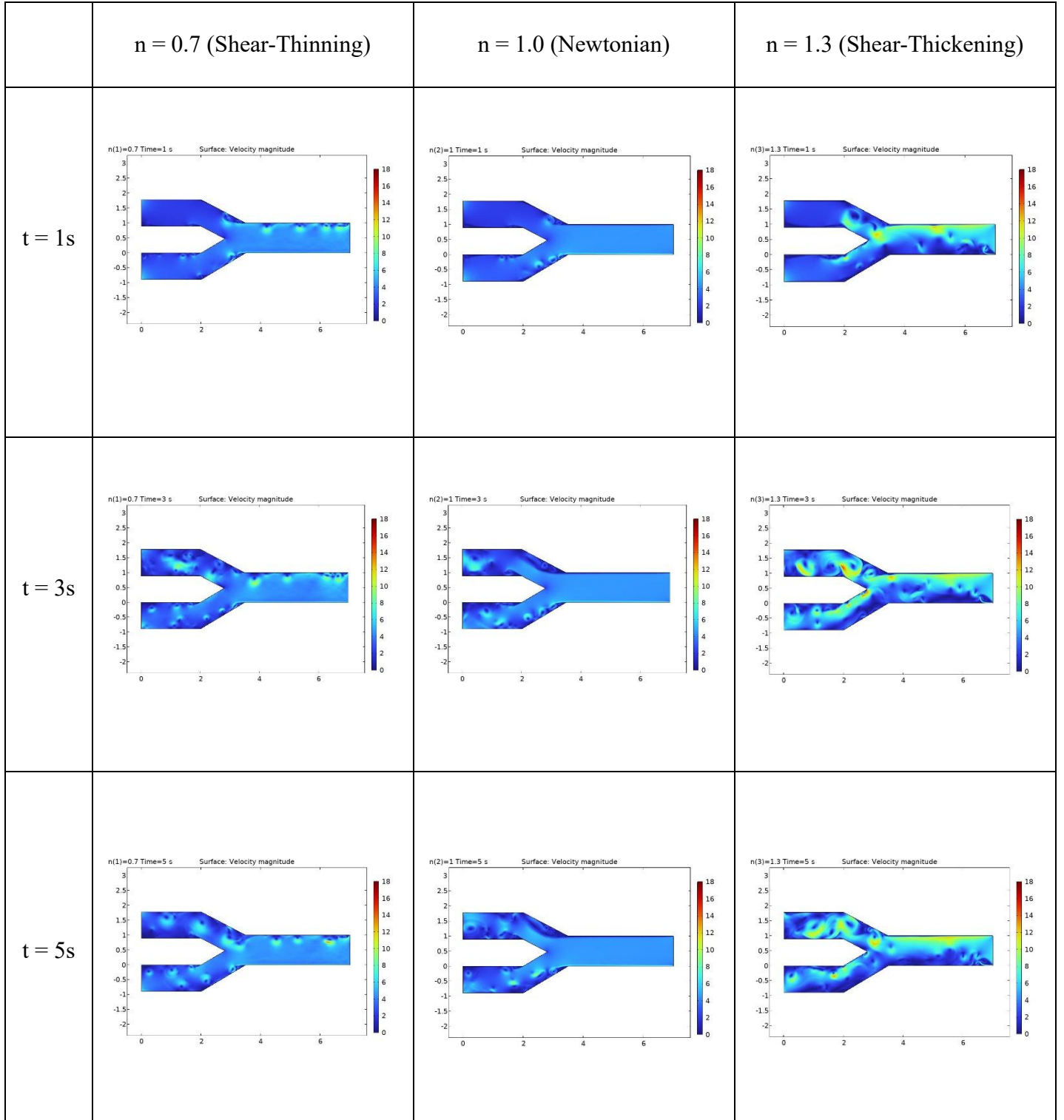


Fig. 8. Spatiotemporal evolution of velocity magnitude contours across different flow behavior indices for $\Delta T = 1\text{ K}$ ($T_w = 311\text{ K}$) across $n = 0.7, 1.0, 1.3$ at $t = 1\text{ s}, 3\text{ s}, 5\text{ s}$.

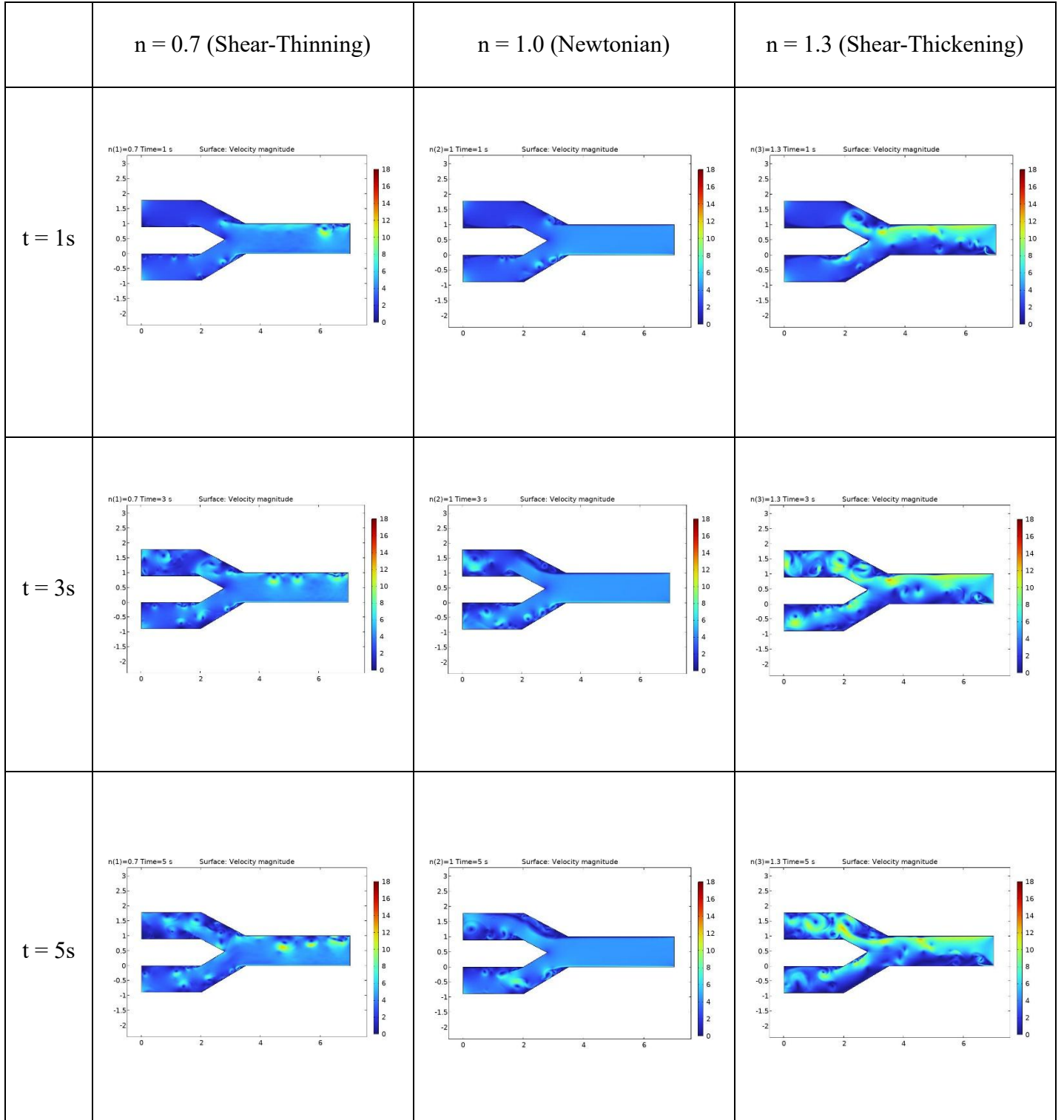


Fig. 9. Spatiotemporal evolution of velocity magnitude contours across different flow behavior indices for $\Delta T = 3\text{ K}$ ($T_w = 313\text{ K}$) across $n = 0.7, 1.0, 1.3$ at $t = 1\text{ s}, 3\text{ s}, 5\text{ s}$.

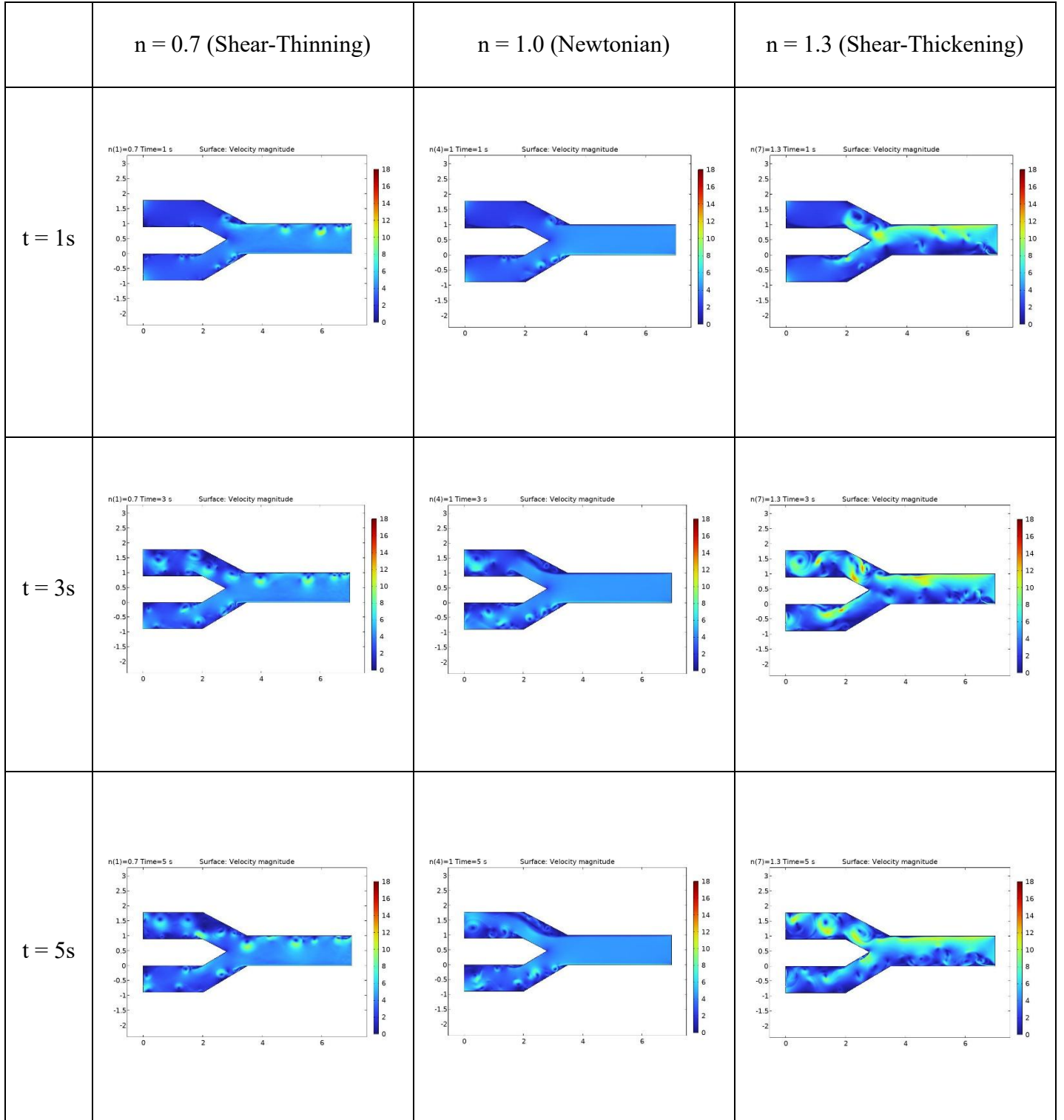


Fig. 10. Spatiotemporal evolution of velocity magnitude contours across different flow behavior indices for $\Delta T = 5\text{ K}$ ($T_w = 315\text{ K}$) across $n = 0.7, 1.0, 1.3$ at $t = 1\text{ s}, 3\text{ s}, 5\text{ s}$.

4. Conclusion

In this investigation, the convective heat transfer performance and non-Newtonian flow behavior of power-law fluids traversing a bifurcating microchannel junction were systematically evaluated using finite element simulations in COMSOL Multiphysics. By assessing parametric combinations of the rheological index ($n = 0.7, 1.0, 1.3$) and heating temperature differentials ($\Delta T = 1 \text{ K}, 3 \text{ K}, 5 \text{ K}$), the study yields the following conclusions:

1. **Convective Thermal Evolution:** Imposed thermal differential ΔT governs the scalar magnitude of heat flux, whereas the internal isotherm topology is regulated by fluid rheology. Shear-thinning fluids ($n = 0.7$) promote convective penetration into the core stream due to enhanced near-wall advection, whereas shear-thickening fluids ($n = 1.3$) exhibit confined thermal boundary layers.
2. **Hydrodynamic Resistance and Pumping Cost:** Increasing the flow index from 0.7 to 1.3 causes a monotonic surge in overall pressure drop across the junction, identifying shear-thinning fluid transport as the most energy-efficient regime for microfluidic operation.
3. **Kinematic Modification:** Pseudoplasticity results in blunt, plug-flow velocity fields, while dilatant rheology accelerates core center-line velocities toward the dividing junction apex.

Future Scope: Subsequent phases of this research will introduce multiphase droplet tracking via Cahn–Hilliard Phase-Field equations to analyze how surface wettability (via contact angle variations $\theta_w = 60^\circ, 90^\circ, 120^\circ$) affects droplet splitting, deformation, and thermal dispersion at the branching divider.

References

1. Akbar, N. S., & Butt, A. W. (2017). Entropy generation analysis for the peristaltic flow of Cu-water nanofluid in a tube with viscous dissipation. *Journal of Hydrodynamics*, 29(1), 135–143. [https://doi.org/10.1016/S1001-6058\(16\)60724-1](https://doi.org/10.1016/S1001-6058(16)60724-1)
2. Tripathi, D., & Bég, O. A. (2014). A study on peristaltic flow of nanofluids: application in drug delivery systems. *International Journal of Heat and Mass Transfer*, 70, 61–70. <https://doi.org/10.1016/j.ijheatmasstransfer.2013.10.044>
3. Kumawat, C., Sharma, B. K., Mekheimer, K. S., Muhammad, T., & Ali, L. (2024). Computer simulation of two phase power-law nanofluid of blood flow through a curved overlapping stenosed artery with induced magnetic field. *International Journal of Numerical Methods for Heat & Fluid Flow*, 34(2), 741–772. <https://doi.org/10.1108/HFF-07-2023-0402>
4. Sain, S., Manna, N. K., Sarkar, S., Santra, S., & Mandal, D. K. (2025). Hemodynamic analysis of Au–Ag hybrid nanofluid flow through curved stenosed arterial geometry. *The European Physical Journal Special Topics*, 1–16. <https://doi.org/10.1140/epjs/s11734-025-02010-2>[cite: 1]
5. Bird, R. B., Armstrong, R. C., & Hassager, O. (1987). *Dynamics of Polymeric Liquids, Volume 1: Fluid Mechanics*. John Wiley & Sons. <https://www.wiley.com/en-us/Dynamics+of+Polymeric+Liquids%2C+Volume+1%3A+Fluid+Mechanics%2C+2nd+Edition-p-9780471802457>
6. Hayat, T., Tanveer, A., & Alsaedi, A. (2016). Mixed convective peristaltic flow of Carreau-Yasuda fluid with thermal deposition and chemical reaction. *International Journal of Heat and Mass Transfer*, 96, 474–481. <https://doi.org/10.1016/j.ijheatmasstransfer.2016.01.054>
7. Sinha, A., Shit, G. C., & Ranjit, N. K. (2015). Peristaltic transport of MHD flow and heat transfer in an asymmetric channel: effects of variable viscosity, velocity-slip and temperature jump. *Alexandria Engineering Journal*, 54(3), 691–704. <https://doi.org/10.1016/j.aej.2015.05.004>
8. Srinivas, S., Gayathri, R., & Kothandapani, M. (2009). The influence of slip conditions, wall properties and heat transfer on MHD peristaltic transport. *Computer Physics Communications*, 180(11), 2115–2122. <https://doi.org/10.1016/j.cpc.2009.06.014>
9. Nejad, A. A., Talebi, Z., Cheraghali, D., Shahbani-Zahiri, A., & Norouzi, M. (2018). Pulsatile flow of non-Newtonian blood fluid inside stenosed arteries: investigating the effects of viscoelastic and elastic walls, arteriosclerosis, and polycythemia diseases. *Computer Methods and Programs in Biomedicine*, 154, 109–122. <https://doi.org/10.1016/j.cmpb.2017.11.013>
10. Gijssen, F. J. H., Allanic, E., van de Vosse, F. N., & Janssen, J. D. (1999). The influence of the non-Newtonian properties of blood on the flow in large arteries.

- Journal of Biomechanics*, 32(7), 705–713. [https://doi.org/10.1016/S0021-9290\(99\)00015-9](https://doi.org/10.1016/S0021-9290(99)00015-9)
11. Chakraborty, S. (2006). Augmentation of peristaltic microflows through electro-osmotic mechanisms. *Journal of Physics D: Applied Physics*, 39(24), 5356–5363. <https://doi.org/10.1088/0022-3727/39/24/034>
 12. Udupa, M. C., Saha, S., & Natarajan, S. (2025). Numerical modeling of non-Newtonian hemodynamic performance and recirculation under resting and exercise conditions. *Scientific Reports*, 15, 1845. <https://doi.org/10.1038/s41598-025-01845-x>
 13. Hussain, A., Sarwar, L., Rehman, A., Akbar, S., Gamaoun, F., Coban, H. H., ... & Alqurashi, M. S. (2022). Heat transfer analysis and effects of (silver and gold) nanoparticles on blood flow inside arterial stenosis. *Applied Sciences*, 12(3), 1601. <https://doi.org/10.3390/app12031601>
 14. Sheikholeslami, M., & Ganji, D. D. (2015). Nanofluid flow and heat transfer between parallel plates considering Brownian motion using DTM. *Computer Methods in Applied Mechanics and Engineering*, 283, 651–663. <https://doi.org/10.1016/j.cma.2014.09.038>